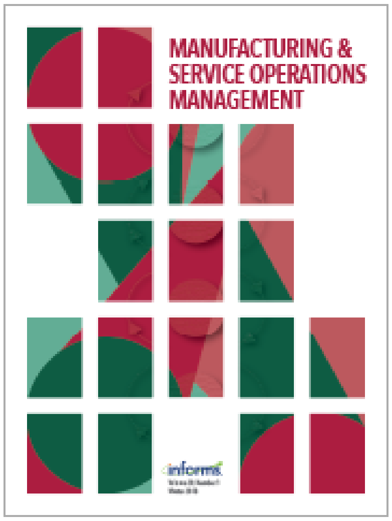

Manufacturing & Service Operations Management

Publication details, including instructions for authors and subscription information:
http://pubsonline.informs.org

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MANUFACTURING & SERVICE OPERATIONS MANAGEMENT
Vol. 27, No. 4, July–August 2025, pp. 993–1007
ISSN 1523-4614 (print), ISSN 1526-5498 (online)
https://pubsonline.informs.org/journal/msom

The High Impact of Hurricanes on Prices in Low-Income Communities

Xabier Barriola,[a,*] **William Schmidt**[b]

[a] School of Economics and Management, Tilburg University, 5037 AB Tilburg, Netherlands; [b] Goizueta Business School, Emory University, Atlanta, Georgia 30322

*Corresponding author

Contact: x.barriola@tilburguniversity.edu, https://orcid.org/0000-0002-0776-4754 (XB); william.schmidt@emory.edu, https://orcid.org/0000-0003-1424-7437 (WS)

Received: June 18, 2023
Revised: June 24, 2024; January 22, 2025
Accepted: April 9, 2025
Published Online in Articles in Advance: April 30, 2025

https://doi.org/10.1287/msom.2023.0337

Abstract. ***Problem definition*:** Responding to large hurricanes requires the deployment of food, shelter, and medical assistance to relieve the immediate needs of the victims. These services are supported by, and eventually fully transitioned to, local resources, such as grocery stores. However, disparities in local infrastructure and services may result in low-income communities bearing a disproportionate burden from these disasters. We investigate this possibility by testing for differences in the percentage change in paid prices, stockouts, and substitutions for grocery products between low-income and high-income communities following three large Atlantic hurricanes. ***Methodology/results*:** We use major disaster declarations by the U.S. federal government, and marry this information to U.S. Census Bureau data on household demographics and third-party data on grocery store sales. Using a triple-difference regression specification, we find that low-income communities in the disaster zones endure higher average percentage price increases within grocery categories compared with high-income communities. We provide evidence for several mechanisms that can contribute to this outcome—higher percent price increases at the product level, more frequent stockouts of low-priced products, and a larger increase in stockout-based substitution from low-priced products to high-priced products. ***Managerial implications*:** Product pricing, stockouts, and stockout-based substitutions are important topics in operations management practice and research. We provide empirical evidence across these outcomes of sustained heterogeneous effects by community income level in the aftermath of a hurricane. These findings can inform private and public sector responses to hurricanes, including pricing, prepositioning target goods, strategic overstocking, grocery store access, and structuring cash-and-voucher (C&V) programs that anticipate the disaster-induced substitution patterns of different community types.

Supplemental Material: The online appendix is available at https://doi.org/10.1287/msom.2023.0337.

Keywords: disaster and humanitarian operations • product substitution • empirical operations

1. Introduction

Millions of people have been affected by natural mega-disasters, such as Hurricane Katrina in 2005, Cyclone Nargis in 2008, and Hurricane Ian in 2022, and weather-related disasters are expected to grow in size and frequency as the Earth warms (Kossin et al. 2020, IPCC 2021). But such events can impose a greater burden on less-advantaged communities. For instance, Deaton (2017) reports that low-income communities are more vulnerable to large weather events because those communities are more likely to suffer from deficient infrastructure, poor drainage, and zoning differences. Collectively, infrastructure differences may make it difficult to provide necessary services to those communities when disaster strikes, including resupplying local grocery stores. In retail settings, these larger burdens may have operational implications, including stockouts, store closures, and restrictions on planned price changes. The objective of our research is to establish whether low-income communities bear disproportionately higher increases in paid prices compared with high-income communities following a hurricane, and describe the operational mechanisms that may contribute to these results.

We examine this issue in the context of retail grocery products following three Atlantic hurricanes—Katrina (2005), Ike (2008), and Sandy (2012). These are the costliest Atlantic hurricanes during our study period—Katrina at \$200B, Ike at \$43B, and Sandy at \$89B, adjusted to 2024 dollars (NOAA 2024). The hurricanes have a wide variation in costs, occurred in different years, and hit different geographies. Conducting our analysis independently for each hurricane provides some insight that our findings are not isolated to one mega-event (Hurricane Katrina) but generalize to other large but less damaging events (Hurricanes Ike and Sandy).

In our analysis, we identify communities at the county level that have been declared a major disaster area by the U.S. federal government following each hurricane. We augment this with detailed grocery store sales data provided by Information Resources, Inc. (IRI), and zip code-level household income and demographic data from the U.S. Census Bureau. The IRI data set is classified into 30 different product categories, spans 12 years, and covers around 200 million observations per year at the product-store-week level. Using these combined data, we estimate a triple-difference regression specification that compares (1) stores in declared disaster areas and a set of unaffected stores, (2) low- and high-income communities, and (3) pre- and post-disaster-declaration periods.

Quantifying the differential impact of hurricanes on grocery products is challenging because the prices paid by consumers can depend not only on product prices but also on changes in paid prices within a category because of stockouts and substitution. The latter effects can dominate, especially following a large disruption such as a hurricane. We address this challenge by conducting analyses at both the product and category levels, and identifying paid price changes within each category. Across the events we study, we find that low-income communities in the disaster zones endure higher average percentage price increases within grocery categories compared with high-income communities. The results indicate an average differential increase in prices paid by low-income communities for grocery products of 2.9% in the 10 weeks following a hurricane, with variation in the effect size by hurricane (4.9% for Katrina, 1.3% for Ike, and 2.2% for Sandy) that is commensurate with the variation in the magnitude of each hurricane's economic damage.

Increases in paid prices may be driven by a variety of underlying mechanisms, including demand changes, larger unit price increases, more stockouts, and greater substitution to high-priced products. Although we cannot prove or disprove all the potential mechanisms, we provide evidence that some of these mechanisms are present in our data and the negative impact falls more heavily on low-income communities, to different degrees. We conjecture that low inventory levels and stockouts may be a common contributing factor to these mechanisms. For instance, store managers may be hesitant to implement planned price reductions on products that unexpectedly stock out or have low inventory following a disruption. Because popular products in low-income communities are often low-priced, stockouts of preferred products may lead those consumers to substitute to higher-priced goods. We provide empirical evidence that posthurricane stockouts of low-priced products are more concentrated in low-income communities, and those communities disproportionately substitute to higher-priced items.

We contribute to the operations management literature on customer heterogeneity, in particular the differential impact of disasters on consumers. Our findings can help operations managers and policymakers better serve communities affected by hurricanes by informing responses such as prepositioning target goods, strategic overstocking, and structuring cash-and-voucher (C&V) programs that anticipate the disaster-induced substitution patterns of different community types. To develop estimates for future events, we establish how to employ readily available grocery store data to examine differential changes in purchasing patterns, stockouts, and substitution patterns across communities. Further research may consider how to refine existing emergency response efforts, including the optimal composition of C&V programs within a larger set of aid resources, more effective design and deployment of voucher programs, and improvements to ordering, shipment, and warehousing decisions. This can help companies better anticipate the needs of the communities that they serve and potentially improve their operational decisions before, during, and after a hurricane.

2. Literature Review

Our research adds to the literature on disaster operations management, which spans critical topics such as efficient debris removal following disasters (Çelik et al. 2015), wartime inventory management (Jola-Sanchez and Serpa 2021), development aid (Sodhi and Knuckles 2021), funding for disaster readiness and response capacity (Natarajan and Swaminathan 2014, Mejia et al. 2019), deployment of disaster medical services (Mills et al. 2018), empirical and analytical support for structuring cash assistance policies during a crisis (Kotsi et al. 2022), the moderating effect of disaster settings on humanitarian project completion (Urrea and Yoo 2023), and prepositioning emergency relief items (Eftekhar et al. 2022, Uichanco 2022). Recent work in humanitarian logistics considers the role of social equity in the provision of various services, including a model to balance equity, efficiency, and effectiveness in food bank operations (Hasnain et al. 2021) and the equitable distribution of pharmaceuticals in sub-Saharan countries (Gallien et al. 2021). Our empirical evidence of community heterogeneity in the pricing and provision of commercial grocery products within a disaster zone extends ongoing work on disaster operations and equity in humanitarian logistics. For instance, accounting for heterogeneous consumption and substitution patterns across communities is as relevant to prepositioning emergency relief items as it is to prepositioning grocery items prior to a disaster. An overview of research on the broader topic of humanitarian operations management can be found in Besiou and Van Wassenhove (2020).

Our research also extends the burgeoning scholarship in operational disruptions. Much of the work in this area focuses on the impact of disruptions on the company's operations (Hendricks and Singhal 2005a, Simchi-Levi et al. 2015), its investors (Hendricks and Singhal 2005b, Schmidt and Raman 2022), or its supply chain (Dolgui and Ivanov 2021). Related work considers how to mitigate the adverse effects of disruptions (Wang et al. 2010, Dong and Tomlin 2012, Mehrotra and Schmidt 2021) and mitigation in the context of small businesses (Kundu et al. 2024). Our research augments this literature by considering how operational disruptions from a hurricane can have a disparate impact on different communities of retail customers. Other research leverages grocery store data, as we do, but to different ends. Pan et al. (2020) use data from a set of physical stores to study the availability of bottled water before and after four hurricanes in the United States. Gagnon and López-Salido (2020) find little evidence for meaningful changes in aggregate average unit prices at grocery stores surrounding demand shocks, including natural disasters. We complement this literature by focusing on paid prices within categories, which account for other effects on purchasing power, including stockouts and product substitutions. We also consider heterogeneous treatment effects and establish that the adverse impacts on unit pricing and stockout-based substitution are concentrated in low-income communities.

Our findings are also relevant to the literature on income-based inequity in product pricing and consumption. Allcott et al. (2019) analyze the prevalence of food deserts in low-income communities. They show that poor areas have more difficulties getting access to fresh food as there are fewer supermarkets compared with richer areas. Broda et al. (2009) study steady-state differences in paid prices between high- and low-income communities and find that low-income households pay less for groceries because they tend to purchase food that is discounted. Handbury (2021) shows a correlation between a community's income level and the assortment of products and prices that stores offer in those communities. Fothergill and Peek (2004) show that hurricane preparedness is related to household income as high-income households have the financial resources to purchase emergency supplies in the face of natural disasters.

Finally, our research is related to retail operations generally, and substitution effects specifically. Pricing, stockouts, assortment planning, and stockout-based substitutions are broad and deep research fields in operations management (for example, Bitran and Caldentey 2003, Anderson et al. 2006, Kök and Fisher 2007, Vulcano et al. 2012, Kök et al. 2015, Chen and Mišić 2022). This research often focuses on settings with well-defined, random demand, rather than settings with large, discrete shocks. Our research augments our theoretical understanding of large shocks by providing empirical evidence that they lead to stockouts, substitution patterns, and paid prices that are heterogeneous across community types. Our contribution establishes that disruption-induced stockouts may systematically and disproportionately disadvantage some consumer classes over others.

3. U.S. Federal Major Disaster Declarations and Food Assistance

Federal, state, and local governments in the United States provide a range of services to households affected by natural disasters, including emergency medical services, property repair, unemployment support, and food assistance. The U.S. federal government established processes allowing states to formally request federal support following natural disasters. In most states, this process starts with a state's Governor issuing an Executive Order declaring a State of Emergency. Local, state, and federal agencies then conduct damage assessments, and the state determines whether those assessments qualify for the Federal Emergency Management Agency's (FEMA's) damage and emergency response assistance. If the qualifications are met, the state's Governor can request that the U.S. President make a federal Major Disaster Declaration. FEMA reviews this request and the supporting evidence, and makes a recommendation to the President, who retains final authority to make a disaster declaration and trigger federal financial and physical assistance.

The insights from our research are most related to the provision of food assistance in federally declared major disasters. FEMA and the U.S. Department of Agriculture (USDA) collectively support three primary types of food assistance following a disaster. The USDA Foods Program provides bulk food to local aid organizations, such as the Red Cross. Those organizations establish congregate feeding locations, prepare the food, and serve community members, particularly those who are homeless, displaced, or without access to electricity. A second type of disaster food assistance, the USDA Disaster Household Distribution Program, provides food boxes to households who can cook their own meals and are sheltering in place. These two programs are focused on meeting immediate nutrition needs in affected communities, and are tapered as local food sources, such as local grocery stores, resume operations. The last program, the Disaster Supplemental Nutrition Assistance Program (D-SNAP), can be implemented if the disaster declaration also includes individual assistance. This provides direct payments through government-issued debit cards to families who are not receiving other forms of disaster food assistance. The debit cards can only be used to purchase food from approved categories, and commercially operated food channels must be open so households have places to redeem their benefits.

These resources do not address the longer-term effects of higher paid prices for grocery products following a disaster. The disaster assistance is intended to be temporary, and in the case of D-SNAP, food must be purchased through local grocery stores. This may lead to reduced purchasing power if fixed benefits are not adjusted to account for rising prices or if stockouts compel households to substitute to higher-priced products. D-SNAP, which has the longest anticipated service period, typically provides one month of benefits to eligible households (USDA 2014), although that period was extended to three months for some of the communities impacted by Hurricane Katrina and two months for some of the communities impacted by Hurricane Ike. We find higher paid price effects that extend well beyond these periods, and extend into non-food-related grocery categories. Households can only qualify for D-SNAP if they do not already receive the maximum amount of food assistance through the USDA Supplemental Nutrition Assistance Program (SNAP), a food nutrition program that serves low-income households.[1] This means that low-income households who already receive SNAP benefits can be excluded from additional benefits through D-SNAP, which limits the incremental benefits flowing to low-income communities.

4. Hypotheses

Inequitable outcomes associated with community income levels can manifest in the aftermath of natural disasters, including regressive allocation of FEMA grants following disasters (Billings et al. 2022) and a higher concentration of federal flood buyouts in low-income communities (Jowers et al. 2023). This can persist even among countries that emphasize sustainability. Besiou et al. (2021) highlight the value of humanitarian operations to address societal problems, but point out that "countries closer to fulfilling the United Nations sustainable development goals (SDGs) suffer less from disasters but still struggle with issues such as social equity" (p. 4343).

Whether a hurricane leads low-income communities to experience a larger increase in paid prices for grocery products compared with high-income communities is an empirical question, but several operational realities hint at its plausibility. Allcott et al. (2019) provide empirical evidence that low-income communities tend to have less access to groceries, especially fresh food. Fothergill and Peek (2004) summarize research on poverty and disasters in the United States that associates low-income communities with lower levels of disaster preparedness and lower-quality construction. Faber and Fally (2022) use detailed home and store product scanner data to show that larger, more productive firms tend to serve high-income households rather than low-income households. These firm characteristics may also allow those firms to provide better service following disruptions, including lower prices and fewer stockouts. Collectively, the factors raised by this stream of research could influence not only the direct impact of disasters on low-income households, but also an indirect impact because of differential resilience in the infrastructure and services that are present in those communities. These observations lead us to propose the following hypothesis.

Hypothesis 1. *Compared with high-income communities, low-income communities experience a larger increase in paid prices for groceries following a hurricane.*

We investigate several operational mechanisms that could shed light on income-based differences in paid prices following a hurricane, including differential changes in demand, unit pricing, and stockout-induced substitution. Section 8.6 provides more details on the scope and data limitations that prevent us from evaluating all of the possible mechanisms that can contribute to differential changes in paid prices. We outline several factors that may contribute to heterogeneous effects by income.

Price changes are typically scheduled to occur in batches in brick-and-mortar retail outlets (DellaVigna and Gentzkow 2019). Given the large number of Universal Product Codes (UPCs) in a store, prices on dozens or even hundreds of UPCs may be changed on any particular week. Because of the prevalence of antigouging laws and the time required to plan and execute price changes, stores may be unwilling to directly raise prices on short notice following a hurricane. They may, however, replace planned price decreases with less aggressive price decreases, while making fewer adjustments to planned price increases. To the extent that such practices are not consistent across all grocery stores, they can contribute to differential pricing changes across communities following a hurricane.

Broda et al. (2009) find that under normal circumstances, low-income households gravitate toward purchasing discounted products, allowing those households to spend less on food than high-income households. The authors do not consider behavior during a disruption. After a disaster, the pattern that Broda et al. (2009) find may lead to lower prices for low-income communities as those communities try to economize more, or it may lead to higher prices in low-income communities if discounted products stock out, experience a supply disruption, or are taken off discount.

Lower-margin products and lower store profits in low-income communities may lead to operational trade-offs, including lower levels of safety stock or less frequent resupply (Porteus 2002). Stores serving low-income communities may be more vulnerable to stockouts as a result. Stores in high-income communities may also be vulnerable to stockouts, potentially for other reasons, including a higher prevalence of stockpiling behavior in the community and larger replacement purchases of storm-damaged products (Fothergill and Peek 2004). Stockouts naturally lead to substitution and prolonged stockouts can lead to persistent substitution (Fitzsimons 2000, Anderson et al. 2006). If stores maintain lower levels of safety stock for lower-priced, and potentially less profitable, goods, it can induce substitution to higher-priced goods. Because popular products in low-income communities are more often low-priced (Handbury 2021), stockouts of preferred products may mean that those consumers must substitute to higher-priced goods whereas high-income communities may be able to substitute to lower-priced products (Brainard 2022). We test for the net result of these effects with the following hypotheses:

Hypothesis 2(a). *Compared with high-income communities, low-income communities experience larger increases in unit prices following a hurricane.*

Hypothesis 2(b). *Compared with high-income communities, low-income communities experience larger stockout-based substitution to high-priced groceries following a hurricane.*

5. Data

Our research leverages grocery store scanner data supplied by Information Resources, Inc. Bronnenberg et al. (2008) provide a detailed description of the data, and describe them as covering markets that correspond roughly to those covered by the U.S. Bureau of Labor Statistics, with the addition of some smaller markets in the IRI data. The IRI data capture detailed weekly price and quantity information for 30 different product categories from over 1,500 grocery stores across the United States. Each grocery store reports weekly information by each product's UPC, including units, dollar sales, descriptions, and brands. IRI also provides information on store geographies, including the zip code, county, and state. The raw data set is an unbalanced panel that runs from January 2001 until December 2012, and the unit of observation is at the UPC-week-store level.

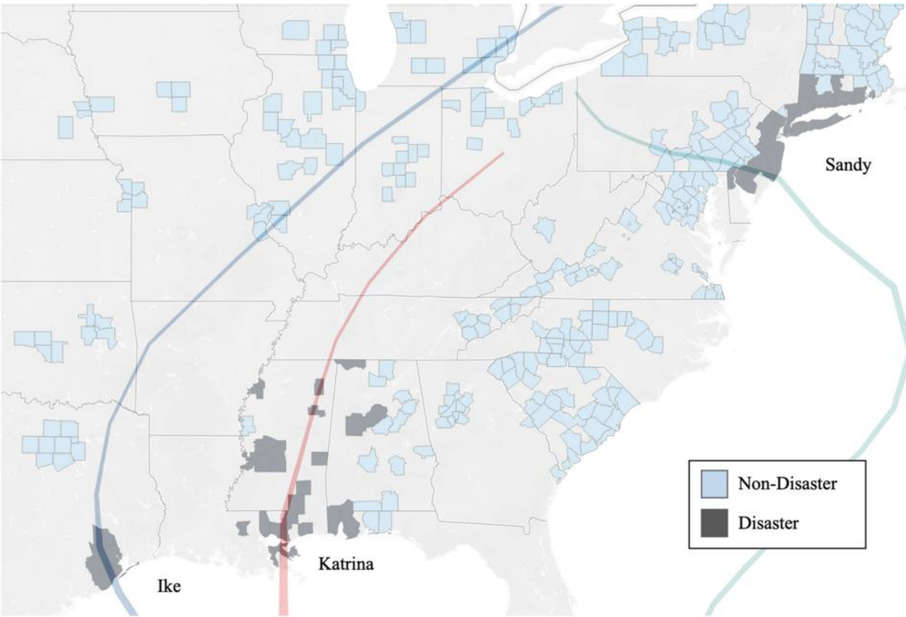
We leverage the IRI data surrounding three major Atlantic hurricanes that impacted different regions of the United States over different time periods: Hurricanes Katrina (2005), Ike (2008), and Sandy (2012).[2] These hurricanes are the three most damaging hurricanes to strike the United States during our analysis period, and the first, fifth, and ninth most damaging hurricanes to hit the United States through 2024 (NOAA 2024). Focusing on large hurricanes provides us with an adequate sample size of affected stores in low- and high-income communities. We identify a community as being materially impacted by a hurricane if the federal government makes a major disaster declaration for the community within three weeks of the hurricane making landfall in the United States. Most disaster declarations occur within days after a hurricane makes landfall. For instance, Hurricane Katrina made its initial landfall on August 25, 2005, in Florida; strengthened over the Gulf of Mexico; and made its second landfall on August 29, 2005, in Louisiana and Mississippi. On that day, FEMA declared Hurricane Katrina to be a major disaster for several counties located in Louisiana (31 parishes), Mississippi (49 counties), and Alabama (11 counties) (FEMA 2021). Figure 1 identifies the path of each hurricane and the counties with stores that are in the IRI data.

Stores in a county that is part of a declared disaster area are identified as treated, and stores that are outside of these counties are candidates for controls. We conduct our analysis using stores that remained open continuously 10 weeks before and at least 4 weeks in the 10 weeks after the hurricanes. This yields 25 treated and 1,153 prospective unaffected stores for Hurricane Katrina, 33 treated and 1,068 prospective unaffected stores for Hurricane Ike, and 84 treated and 793 prospective unaffected stores for Hurricane Sandy.

We join our grocery store data with household demographic details provided at the zip code level through the U.S. Census Bureau. For Hurricanes Katrina and Ike, we use the 2000 U.S. Census data for each zip code to extract median household income, the proportion of the community who identify as Black, and the proportion of the community with a college degree. Median income is not captured in the 2010 Census at the zip code level, so for Hurricane Sandy, we use data from the Census Bureau's 2011 American Community Survey. From these two sources, we create community variables that are assigned to stores located within each zip code. *College Degree* identifies the proportion of the population in the zip code who have a college degree. *Black* identifies the proportion of the population in the zip code who self-identify as Black. *College Degree* and *Black* are used in robustness checks described in the Online Appendix. *Income Quartile* identifies the quartile for each zip code based on the median incomes over all zip codes represented in the IRI data within a state.

We generate three variables to serve as the basis for our triple-difference regression specification. We construct *Path* as an indicator variable that is equal to "1" for stores located in a declared disaster area, and "0" otherwise. *Post* is an indicator variable that is equal to "1" in periods after the hurricane, and "0" in periods prior to the hurricane. *Low* is an indicator variable that is set to "1" if the store is located in a zip code that is in the bottom quartile based on *Income Quartile*, and "0" if it is

Figure 1. (Color online) Paths of Hurricanes Katrina, Ike, and Sandy and Communities with IRI Data



Notes. The lines represent the paths of Hurricanes Katrina, Ike, and Sandy. The counties with stores reporting IRI data are identified with shading that is either dark (representing disaster-declared counties) or light (representing nondisaster counties).

in the top quartile. Table 1 provides income information for the disaster communities for each of the three hurricanes. As this table indicates, predisaster income levels for the affected communities differ materially by hurricane. Income levels also vary materially across states, reinforcing the need to use within-state income quartiles to disambiguate high- and low-income communities.

Our analysis focuses on comparing the paid prices in stores located in the top and bottom income quartiles. After dropping observations in the second and third income quartiles, we are left with 11 treated and 564 prospective unaffected stores for Hurricane Katrina, 21 treated and 519 prospective unaffected stores for Hurricane Ike, and 44 treated and 397 prospective unaffected stores for Hurricane Sandy. Table OA1 of the Online Appendix summarizes the exclusions for the treated stores used for each hurricane. In robustness checks, we show that our results are consistent if we instead assess all four income quartiles, identify *Low* using the bottom and top income quintiles, or identify *Low* using below versus above the median income.

The set of variables in our analysis can be estimated at different levels of detail, including different combinations of store (s), week (w), category (c), and product (p). We use these subscripts to represent the level of detail for a variable in each analysis unless the meaning is otherwise clear. *Units* identifies the total number of units sold, *Dollars* identifies the total dollars spent, *Assortment* identifies the number of unique products (also identified as UPCs), and *Price* identifies the average price paid (calculated as *Dollars* divided by *Units*), all at the defined level of detail. For example, the average price paid at the category, store, week level of detail is represented as $Price_{csw}$. For many of our analyses, we take the natural logarithm of the variable of interest. This is represented by prepending "*Log*" to the variable name, for example, $LogPrice_{csw}$.

We run our base-case analyses for each hurricane using 10 weeks before and 10 weeks after each of the three hurricanes. We define the postperiod as starting in the week that the local hurricane warning is announced. For each of the hurricanes we study, the hurricanes

Table 1. Distribution of Income for Disaster Communities

	Katrina			Ike			Sandy		
	Mean	Median	SD	Mean	Median	SD	Mean	Median	SD
High-income (Low = 0)	50,114	47,791	4,684	76,558	70,587	10,580	115,587	109,343	29,217
Low-income (Low = 1)	25,224	25,651	1,491	32,342	35,524	4,741	54,421	55,343	10,881

Notes. A zip code is identified as low-income (high-income) if it is in the bottom (top) quartile based on *Income Quartile*. All the amounts are rounded to the nearest USD.

Table 2. Hurricane Timelines and Study Periods

Name	Warning announced	Landfall	Preperiod	Postperiod
Katrina	27-Aug-05	29-Aug-05	13-Jun-05 – 21-Aug-05	22-Aug-05 – 30-Oct-05
Ike	11-Sep-08	13-Sep-08	30-Jun-08 – 7-Sep-08	8-Sep-08 – 16-Nov-08
Sandy	27-Oct-12	29-Oct-12	13-Aug-12 – 21-Oct-12	22-Oct-12 – 30-Dec-12

Note. Dates shown as day-month-year.

occur in the latter half of the week, meaning our treatment period starts several days before the hurricane warning is issued. Table 2 summarizes the timelines and study periods for each hurricane. In the Online Appendix, Table OA2 provides store summary statistics and Table OA3 provides community summary statistics. Correlation tables are available from the authors.

6. Empirical Analysis

6.1. Empirical Modeling

The prices paid by consumers can depend on changes in product prices, as well as changes in paid prices within a category because of hurricane-induced spending changes, stockouts, and substitution. To account for this joint effect, we conduct our primary analyses at the category level. Conducting analyses at the category level is common in research on substitution (Kök et al. 2015, Jagabathula and Rusmevichientong 2019, Chen and Mišić 2022) and related research topics, including category aggregations to estimate demand patterns under substitution (Vulcano et al. 2012), promotion optimization considering cross-item substitution and complementarity effects within grocery categories (Cohen et al. 2021), and relationships among grocery category pricing, promotion, and consumer search (Haviv 2022). Russell and Petersen (2000) find little evidence for considering substitution at a level higher than categories by showing minimal support for cross-category price elasticities.

To test Hypothesis 1, we seek to isolate the impact on low-income communities (compared with high-income communities), affected by hurricanes (compared with unaffected), and following those hurricanes (compared with preceding those hurricanes). To achieve this, we employ a triple-difference regression specification along those dimensions. The nondisaster stores in our sample serve as a benchmark of how paid prices at stores in low-income and high-income disaster communities would have differed before and after the disaster declaration date had a hurricane not hit those communities. Our triple-difference regression specification subtracts this benchmark from the observed differences in paid prices at affected stores in low-income communities versus high-income communities, before and after the event of interest. We estimate the following specification separately for each hurricane:

$$\begin{aligned} LogPrice_{csw} = {} & \beta_0 + \beta_1 Path_s + \beta_2 Post_w + \beta_3 Low_s + \beta_4 Path_s \\ & \times Post_w + \beta_5 Path_s \times Low_s + \beta_6 Post_w \times Low_s \\ & + \beta_7 Path_s \times Post_w \times Low_s + \alpha_c + \gamma_s + \delta_w \\ & + \epsilon_{csw}. \end{aligned} \tag{1}$$

The unit of analysis for this model is represented at the category-store-week level. $LogPrice_{csw}$ is the natural logarithm of the weighted average price paid for products in category c, store s, during week w. The log transformation allows us to assess the impact of the coefficients as the percentage change in paid prices. This is relevant in our setting because households in high- and low-income communities spend materially different amounts on groceries absent a disaster (Bureau of Labor Statistics 2007), and we are interested in understanding the relative change in paid prices with respect to each community's normal spending patterns.

We include category fixed effects (α_c), store fixed effects (γ_s), and week fixed effects (δ_w) to control for potential omitted variables that are invariant over category, store, or time. As a consequence, $Path_s$, Low_s, and $Path_s \times Low_s$ are absorbed by the store fixed effects, and $Post_w$ is absorbed by the week fixed effects in our results. The coefficient on $Path_s \times Post_w$ measures the percentage difference in paid prices in affected high-income areas after the event. The coefficient on $Post_w \times Low_s$ measures the percentage difference in paid prices in unaffected low-income areas after the event. The coefficient on $Path_s \times Post_w \times Low_s$ is the triple-difference estimator, which isolates the differential impact on paid prices at a store in a low-income community versus a high-income community, comparing affected stores versus unaffected stores, and in the postdisaster period relative to the predisaster period. For ease of interpretation, we provide information on relevant linear combinations of the coefficients for these terms in our presented results.

Our treatment assignment is determined by federal major disaster declarations, which are at the county level for the events that we study. This leads us to use robust standard errors clustered by county. Abadie et al. (2023) provide evidence that clustering standard errors at the treatment level is necessary when the treatment level differs from the unit of analysis (store-category or store-product, in our analyses). In robustness checks (available from the authors), we confirm that our inferences are consistent if we utilize alternative levels of clustering—store, grocery chain, zip code, geographic proximity, or state. A necessary requirement for our analysis is that the trends of the dependent variable for the treated and control observations are parallel prior to the disaster period for both low-income and high-income observations. In Section OA2.2.2 of the Online Appendix, we provide the results establishing that the parallel trends assumption holds, including the more rigorous equivalence test suggested by Dette and Schumann (2024).

7. Results

7.1. Model-Free Evidence

We start by providing model-free evidence for our main results. This helps to establish that our results are not an artifact of the subsequent modeling choices we employ. Figure 2 presents scaled prices for each hurricane using average prices paid and 95% confidence intervals across all stores, broken down by treated and control stores, and with treated stores further broken down by those serving low-income and high-income communities. To ease comparisons over time and across hurricanes, we scale each price series so that its geometric mean in the 10-week preperiod is equal to one. The panels on the left side (Figure 2, (a), (c), and (e)) include all four income quartiles, and plot the evolution of prices around each hurricane for stores in the disaster area versus outside the disaster area. The panels on the right side (Figure 2, (b), (d), and (f)) plot the evolution of prices around each hurricane for the disaster area stores in low-income versus high-income communities. The price index in the preperiod (weeks 1 through 10) is reasonably stable in each subfigure. For Figure 2, (a), (c), and (e), in the postperiod (weeks 11 through 20), there are larger price increases in disaster area stores compared with the control stores in the aftermath of a hurricane. For Figure 2, (b), (d), and (f), in the postperiod, the increase in the price index is demonstrably larger in low-income communities compared with high-income communities in the aftermath of a hurricane.

7.2. Empirical Evidence

It is useful to identify the average treatment effect of hurricanes on paid prices across all communities. This can be done by dropping *Low* from Equation (1) and constructing a difference-in-differences specification:

$$LogPrice_{csw} = \beta_0 + \beta_1 Path_s + \beta_2 Post_w + \beta_3 Path_s \times Post_w + \alpha_c + \gamma_s + \delta_w + \epsilon_{csw}. \quad (2)$$

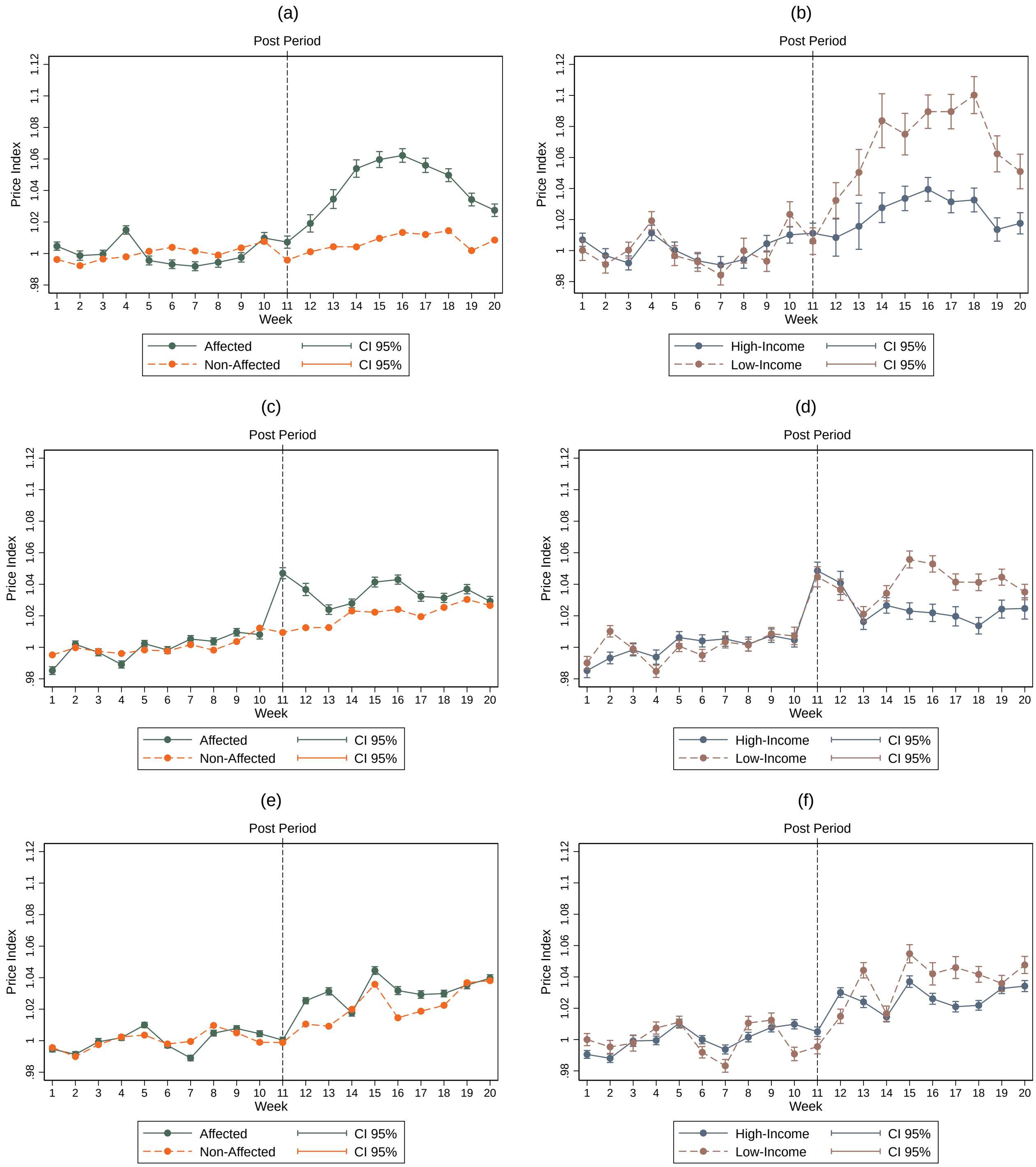
Table 3 summarizes the results from our estimation of Equation (2). The coefficient on *Path* × *Post* is statistically significant and ranges in magnitude between 0.0127 and 0.0263, indicating a 1.3%–2.7% average increase in paid prices for all disaster communities across the three hurricanes.

We now turn to disambiguating the effects by community income level. Table 4 summarizes the heterogeneous treatment effect from our estimation of Equation (1). Across the three hurricanes, the coefficient on *Path* × *Post* summarizes the impact on high-income communities. The value ranges from 0.4% to 0.5%, and is statistically insignificant for all the events. In contrast, the coefficients on *Path* × *Post* × *Low* have a larger magnitude from 1.3% to 5.0%, and are statistically significant for all three hurricanes. This isolates the incremental impact on in-path, low-income communities (relative to high-income communities) following a hurricane, and provides strong support for Hypothesis 1.

Other ways of presenting the impact of hurricanes on the paid prices in low-income communities yield similar insights. The linear combination of *Path* × *Post* and *Path* × *Post* × *Low* represents the incremental impact on in-path, low-income communities in the postperiod, compared with out-of-path, low-income communities in the postperiod. There is a differential price increase for in-path low-income communities of 5.4% following Katrina, 1.9% following Ike, and 2.7% following Sandy. We also present the linear combination of *Path* × *Post*, *Post* × *Low*, and *Path* × *Post* × *Low*, which represents the incremental impact on in-path, low-income communities in the postperiod, compared with in-path, low-income communities in the preperiod. This average effect is a price increase for in-path, low-income communities of 5.3% following Katrina, 1.7% following Ike, and 2.3% following Sandy. In summary, low-income communities experience an increase in paid prices following hurricanes, and the percentage increase in paid prices is materially larger than those experienced by out-of-path, low-income communities or in-path, high-income communities.

To put these incremental impacts into perspective, the U.S. Bureau of Labor Statistics estimates that in 2005, U.S. households in the bottom income quintile paid on average 20.5% of their pretax income on food that is consumed at home, compared with 3.4% for the top-quintile households (Bureau of Labor Statistics 2007). The paid price increase of 5.3% for low-income households following Katrina implies that they would have to spend 21.5% of their pretax income on food, representing a loss in total purchasing power of 1.07 percentage points, compared with a loss of 0.01 percentage points for the top-quintile households. This differential impact on the purchasing power of low-income households is 88 times larger than that on high-income households, and can have a material impact on households that are already financially constrained. This estimate accounts for food groceries only, and underestimates the total impact on the purchasing power of low-income communities as it does not include spending on nonfood

Figure 2. (Color online) The Impact of Hurricanes on Category Average Price



Notes. (a) Treated and control (Katrina); (b) treated high-income vs. low-income (Katrina); (c) treated and control (Ike); (d) treated high-income vs. low-income (Ike); (e) treated and control (Sandy); (f) treated high-income vs. low-income (Sandy). CI, confidence interval.

grocery categories, which represent 14 out of the 30 categories in our study.

7.3. Validity and Robustness Checks

We execute alternative analyses to check whether our results hinge on our underlying assumptions. To conserve space, we present the results of those checks in Section OA2 of the Online Appendix. Section OA2.1 presents the results of a placebo test in which we randomly assign treatment status to stores that are not impacted by a hurricane. Section OA2.2 tests our empirical modeling assumptions, specifically an equivalence

Table 3. The Effect of Hurricanes on Category-Level Paid Prices (DD Regression)

	Katrina (1)	Ike (2)	Sandy (3)
Path × *Post*	0.0263***	0.0127**	0.0139***
	(0.0087)	(0.0050)	(0.0044)
Constant	1.0732***	1.2109***	1.2910***
	(0.0001)	(0.0001)	(0.0002)
Observations	340,805	321,074	249,640
R^2	0.9028	0.8955	0.9052
Unaffected stores	564	519	397
Affected stores	11	21	44

Notes. Robust errors clustered by county are shown in parentheses. Observations are at the category/week/store level. All the models include category, week, and store fixed effects. We use a 10-week preperiod and a 10-week postperiod.
$^*p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

test for parallel trends and a robustness check that guards against cross-lag contamination in a differences-in-differences specification (Goodman-Bacon 2021). Section OA2.3 provides robustness checks that establish a refined set of control observations for our analyses based on trimming observations or employing different matching algorithms. Section OA2.4 presents robustness checks that vary the composition of the sample. This includes analyzing all four income quartiles, defining high and low-income communities using either the median or the top and bottom quintiles, employing different postperiod durations, and conducting our analysis individually for each treated store. The results from all of the validity and robustness checks support our main findings.

Table 4. The Effect of Hurricanes on Category-Level Paid Prices

	Katrina (1)	Ike (2)	Sandy (3)
Path × *Post*	0.0036	0.0052	0.0051
	(0.0055)	(0.0035)	(0.0051)
Post × *Low*	−0.0014	−0.0017	−0.0034**
	(0.0014)	(0.0018)	(0.0015)
Path × *Post* × *Low*	0.0490***	0.0134***	0.0211**
	(0.0096)	(0.0032)	(0.0082)
Constant	1.0736***	1.2113***	1.2920***
	(0.0004)	(0.0005)	(0.0005)
Observations	340,805	321,074	249,640
R^2	0.9028	0.8955	0.9052
Unaffected stores	564	519	397
Affected stores	11	21	44
PP + PPL	0.0526***	0.0186***	0.0263***
PP + PL + PPL	0.0512***	0.0169***	0.0229***

Notes. Robust errors clustered by county are shown in parentheses. Observations are at the category/week/store level. All the models include category, week, and store fixed effects. PP + PPL is the linear combination of *Path* × *Post* and *Path* × *Post* × *Low*. PP + PL + PPL is the linear combination of *Path* × *Post*, *Post* × *Low*, and *Path* × *Post* × *Low*.
$^*p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

8. Operational Mechanisms

We consider several operational mechanisms that may be contributing to our results. The direct mechanism is, of course, the hurricanes. There is little room for direct interventions against the occurrence of hurricanes, however, because they remain stubbornly resistant to human control. Instead, we provide insights on operational mechanisms that can be influenced by practitioners and policymakers and may shed light on why hurricanes disproportionately impact prices paid in low-income communities. This analysis is constrained by the structure of our data. In particular, IRI provides summary sales data (as opposed to demand information and inventory positions) that are aggregated at the week, product, store level (as opposed to the transaction level).

The differential impact on low-income and high-income communities is likely the net effect of many supply-side and demand-side mechanisms. These mechanisms may be present at varying levels across the many stores in our sample, so our test of Hypothesis 1 effectively captures the pooled average effect that results from many possible secondary mechanisms. In this section, we provide support for multiple mechanisms, and by extension, Hypotheses 2(a) and 2(b). In Section 8.6 we emphasize the limits of our analysis.

8.1. Product Price Changes

To assess whether product price increases are driving our results, we use Equation (1) to estimate $LogPrice_{psw}$

Table 5. The Effect of Hurricanes on Product-Level Paid Prices

	Katrina (1)	Ike (2)	Sandy (3)
Path × *Post*	0.0054	0.0029	0.0056
	(0.0037)	(0.0026)	(0.0045)
Post × *Low*	0.0001	−0.0002	−0.0030**
	(0.0013)	(0.0016)	(0.0012)
Path × *Post* × *Low*	0.0257***	0.0019	0.0150**
	(0.0049)	(0.0021)	(0.0066)
Constant	0.9470***	1.0670***	1.1465***
	(0.0003)	(0.0004)	(0.0004)
Observations	32,505,950	35,460,286	28,189,578
R^2	0.9361	0.9430	0.9442
Unaffected stores	564	519	397
Affected stores	11	21	44
PP + PPL	0.0311***	0.00482**	0.0205***
PP + PL + PPL	0.0312***	0.00465*	0.0175***

Notes. Robust errors clustered by county are shown in parentheses. Observations are at the product/week/store level. All the models include product, week, and store fixed effects. PP + PPL is the linear combination of *Path* × *Post* and *Path* × *Post* × *Low*. PP + PL + PPL is the linear combination of *Path* × *Post*, *Post* × *Low*, and *Path* × *Post* × *Low*.
$^*p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

at the product-store-week level and incorporate product rather than category fixed effects. Table 5 summarizes the results. The coefficients on $Path \times Post$ are positive and statistically insignificant for each hurricane, whereas the coefficients on $Path \times Post \times Low$ are positive and statistically significant for Hurricanes Katrina and Sandy. The linear combinations of $Path \times Post$, $Post \times Low$, and $Path \times Post \times Low$ are positive and statistically significant for each hurricane. Collectively, these results provide support for Hypothesis 2(a). However, the product price increases do not fully account for the category price increases identified in Table 4.

8.2. Product Stockouts

Hurricanes can lead to supply disruptions and demand surges, raising the possibility that stockouts may contribute to our category results. Our transaction-based data do not provide information on unmet demand or store inventory levels. However, we can gain some insight into the occurrence of stockouts by examining sales patterns of products that are regularly sold prior to our study period. We define regular sellers for each store as the products that are sold in every week at that store over the five weeks prior to our study period. We define a stockout as a week of no sales in a store for a product that is typically a regular seller. Using this measure as our dependent variable, we run the following fixed effects logit regression at the product/week/store level,

$$\begin{aligned} Stockout_{psw} = \beta_0 &+ \beta_1 Post_w + \beta_2 Low_s + \beta_3 LowPrice_{ps} \\ &+ \beta_4 Post_w \times Low_s + \beta_5 Post_w \times LowPrice_{ps} \\ &+ \beta_6 Low_s \times LowPrice_{ps} + \beta_7 Post_w \times Low_s \\ &\times LowPrice_{ps} + \alpha_p + \gamma_s + \delta_w + \epsilon_{psw}. \end{aligned} \tag{3}$$

$Stockout_{psw}$ is equal to "1" when there are zero units sold of a previously regularly selling product p in store s during week w and "0" otherwise. $LowPrice_{ps}$ is equal to "1" if product p's price is below the median price in its category c and store s over the five weeks before our study period, and "0" if its price is above the median. Low_s and $Post_w$ are as previously defined. We include product, store, and week fixed effects.

The results are presented in Table 6. The linear combination of the coefficients on $Post \times Low$, $Post \times LowPrice$, and $Post \times Low \times LowPrice$ provides evidence that in the postperiod, low-income communities experience a material and statistically significant increase in stockouts of low-priced products following hurricanes. The coefficients on $Post \times Low \times LowPrice$ indicate that the increase in stockouts of low-priced goods was materially and statistically significantly larger for low-income communities compared with high-income communities. We build on this outcome in the next two subsections by analyzing whether disaster areas experienced a shift from low-priced products to higher-priced products and whether the pattern of product stockouts can predict a differential increase in the prices paid for groceries in low-income compared with high-income communities.

Table 6. Stockouts of Low-Priced Goods

	Katrina (1)	Ike (2)	Sandy (3)
LowPrice	−0.0206	0.0046	−0.0107
	(0.0363)	(0.0424)	(0.0184)
Low × *LowPrice*	−0.207***	−0.271***	−0.243***
	(0.0452)	(0.0225)	(0.0213)
Post × *Low*	0.1388*	0.265*	−0.0456
	(0.0703)	(0.134)	(0.0347)
Post × *LowPrice*	0.0653*	0.0305	0.0722***
	(0.0257)	(0.0405)	(0.0195)
Post × *Low* × *LowPrice*	0.159**	0.0612**	0.110***
	(0.0489)	(0.0242)	(0.0234)
Observations	274,033	803,384	1,608,187
Pseudo-R^2	0.237	0.289	0.236
Affected stores	11	21	44
PxLP + PxLxLP	0.224***	0.0918***	0.182***
PxL + PxLP + PxLxLP	0.363***	0.357***	0.137***

Notes. Robust errors clustered by county are shown in parentheses. PxLP + PxLxLP is the linear combination of $Post \times Low$, $Post \times LowPrice$, and $Post \times Low \times LowPrice$. PxL + PxLP + PxLxLP is the linear combination of $Post \times Low$, $Post \times LowPrice$, and $Post \times Low \times LowPrice$.
*$p < 0.10$; **$p < 0.05$; ***$p < 0.01$.

8.3. Purchasing Pattern Changes

We next determine whether low-income communities must reduce their purchases of lower-priced goods following a hurricane. To achieve this, we estimate the natural logarithm of the number of units purchased for each combination of product-store-week, and we use it as a dependent variable for Equation (3).

The results are presented in Table 7. The coefficients on *LowPrice* and $LowPrice \times Low$ intuitively imply that in the preperiod both high- and low-income communities tend to purchase more units of low-priced products compared with high-priced products, and this pattern is more pronounced for low-income communities. The coefficient on $Post \times LowPrice$ indicates that high- and low-income communities reduce the proportion of low-priced products that they purchase following Hurricane Sandy, whereas the coefficient on $Post \times Low \times LowPrice$ provides evidence that low-income communities differentially decrease the proportion of low-priced products that they purchase following Hurricanes Katrina and Ike. These results indicate that communities can transition away from lower-priced products following a hurricane, and this transition is larger and more pronounced for low-income communities.

8.4. Estimating the Impact of Stockouts on Paid Prices

We train a model to estimate the effect of stockouts on prices using data in the 10 weeks that precede our study

Table 7. The Effect of Hurricanes on Purchasing Patterns

	Katrina (1)	Ike (2)	Sandy (3)
LowPrice	0.3947***	0.2773***	0.5540***
	(0.0279)	(0.0325)	(0.0456)
Low × *LowPrice*	0.1166**	0.2042***	0.1755***
	(0.0424)	(0.0295)	(0.0332)
Post × *Low*	0.0010	−0.0306	−0.0204
	(0.1082)	(0.0147)	(0.0129)
Post × *LowPrice*	0.0127	0.0032	−0.0295***
	(0.0143)	(0.0045)	(0.0068)
Post × *Low* × *LowPrice*	−0.0554*	−0.0337*	−0.0079
	(0.0279)	(0.0125)	(0.0105)
Constant	1.1915***	1.2770***	1.1902***
	(0.0184)	(0.0123)	(0.0177)
Observations	502,677	1,330,348	2,696,712
R^2	0.6166	0.6123	0.5632
Affected stores	11	21	44

Notes. Robust errors clustered by county are shown in parentheses. Observations are at the product/week/store level. All the models include product, week, and store fixed effects.
$^*p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

period.[3] To estimate the impact on $LogPrice_{csw}$, we create $StockoutLowPrice_{csw}$ as the number of stockouts for low-priced, regularly selling products in each category, store, week, and $StockoutHighPrice_{csw}$ as the number of stockouts for high-priced, regularly selling products in each category, store, week. To establish whether stockouts can be used to predict paid prices, we train the following prediction model using the hurricane-affected stores:[4]

$$\begin{aligned} LogPrice_{csw} = {} & \beta_0 + \beta_1 StockoutLowPrice_{csw} \\ & + \beta_2 StockoutLowPrice_{csw} \times Low_s \\ & + \beta_3 StockoutHighPrice \\ & + \beta_4 StockoutHighPrice_{csw} \times Low_s + \alpha_c + \gamma_s \\ & + \delta_w + \omega_{csw}. \end{aligned} \tag{4}$$

For prediction models, statistical significance on individual variables is less critical than the statistical significance of the model's predictions. We present the results in Table 8. The coefficients on both *StockoutLowPrice* and *StockoutHighPrice* are small and sometimes negative, indicating a small impact on paid prices in high-income communities because of stockouts. However, the coefficients on the linear combination of *StockoutLowPrice* + *StockoutLowPrice* × *Low* and the linear combination of *StockoutHighPrice* + *StockoutHighPrice* × *Low* indicate that low-income communities consistently pay more when they have to substitute away from their preferred products.

The net effect on paid prices depends on the combination of changes in paid prices because of stockouts of low-priced products and changes in paid prices because of the stockouts of high-priced products. To estimate this net effect, we use the number of high-priced and low-priced stockouts per category/week/store and the

Table 8. The Effect of Stockouts on Category-Level Paid Prices

	Katrina (1)	Ike (2)	Sandy (3)
StockoutLowPrice	0.0008	−0.0004	−0.0011
	(0.0017)	(0.0012)	(0.0008)
Low × *StockoutLowPrice*	0.0079	0.0023	0.0023
	(0.0037)	(0.0026)	(0.0016)
StockoutHighPrice	0.0022	−0.0073	−0.0015
	(0.0027)	(0.0030)	(0.0012)
Low × *StockoutHighPrice*	−0.0003	0.0094	0.0017
	(0.0023)	(0.0051)	(0.0022)
Constant	0.9517	1.1038	1.3010
	(0.0043)	(0.0022)	(0.0019)
Observations	2,590	5,981	11,667
R^2	0.9423	0.9350	0.8909
Affected stores	10	21	44
SLP + SLPxL	0.00866	0.00188	0.00116
SHP + SHPxL	0.00188	0.00220	0.000229

Notes. Robust errors clustered by county are shown in parentheses. Observations are at the category/week/store level. All the models include category, week, and store fixed effects. SLP + SLPxL is the linear combination of *StockoutLowPrice* and *StockoutLowPrice* × *Low*. SHP + SHPxL is the linear combination of *StockoutHighPrice* and *StockoutHighPrice* × *Low*. We use data from 10 weeks before our main study period to create this table.

prediction model weights from Table 8 to determine whether stockouts predict an incremental increase in paid prices in the postperiod for low-income communities compared with high-income communities. Following this process, we estimate that stockouts predict an incremental increase in the paid prices for low-income communities of 0.94 percentage points for Katrina, 1.87 percentage points for Ike, and 0.61 percentage points for Sandy. All three estimates are significant at the 1% level. This provides support for Hypothesis 2(b).

8.5. Other Mechanisms and Extensions

We investigate a possible demand-side mechanisms in Section OA3 of the Online Appendix. In particular, we adjust the study period to guard against temporary evacuations, which may create a demand-side shock that differentially affects low- versus high-income communities. Our insights are unaffected with this adjustment. This analysis was executed primarily in response to suggestions made by anonymous members of the review team and is therefore not hypothesized by the authors. We are grateful for the review team's input.

We present the results of additional extensions to our analysis in Section OA4 of the Online Appendix. In Section OA4.1 we evaluate possible substitution paths for low-income and high-income communities following the data preprocessing steps described in Jagabathula and Rusmevichientong (2019) and find that low- and high-income communities can substitute to higher-priced products, but the effects are more pronounced for low-income communities. In Section OA4.2 we evaluate the number of units sold in grocery stores and provide some evidence that high-income communities increase their purchases following disasters, but the effect is weakened for low-income communities.

8.6. Limitations Assessing Supply and Demand

Our analysis on mechanisms is constrained by the scope and structure of the available data. In particular, IRI's weekly product transaction data are aggregated across customers and the data lack information on product inventory levels, stockouts, and demand. These limitations leave us to use proxies to estimate the occurrence of stockouts, substitution patterns, and demand changes. Despite these limitations, we endeavored to rigorously test for the presence of both supply-side and demand-side mechanisms that could shed light on our main results. We cannot definitively resolve all the mechanisms that may be leading to our results. Although we cannot produce evidence supporting demand-side factors, it does not mean that demand-side factors are not relevant. Our results may be refined by researchers with access to customer transaction histories, inventory information, better demand proxies, or more detailed information on community-level food and support services beyond D-SNAP.

9. Discussion and Implications

Our research contributes to the academic and practical understanding of the heterogeneous impacts of large disasters on important operations metrics relevant to consumers—paid prices, unit prices, and stockouts. Our findings are important for operations managers and policymakers who seek to more equitably serve communities that are affected by hurricanes and can inform governmental and private-sector responses. This includes prepositioning target goods, strategic overstocking, and structuring C&V programs that anticipate the disaster-induced substitution patterns of different community types. We analyze the impact of three large hurricanes on the prices paid for groceries in low-income versus high-income communities, and the operational mechanisms behind the impact. Our analysis reveals that disparities in paid prices exist and disproportionately disadvantage low-income communities. Across these hurricanes, we find evidence that the larger impact on low-income communities can be partially attributable to a mix of product price changes and stockout-based substitution to higher-priced products.

A conceptual contribution of our analysis is to acknowledge and measure the heterogeneous effects of disruptions across consumer classes. The percentage increase in grocery prices following a disruption can be thought of as a form of targeted price inflation. Policymakers recognize that the effects of price inflation can be harder on low-income communities, but point out that such disparities are rarely quantified. This recognition is well summarized in recent comments by the then-Federal Reserve Vice Chair, Lael Brainard, in the context of general inflation: "While national data do not directly disaggregate the differential effects of inflation by household income groups, a variety of evidence suggests that lower-income households disproportionately feel the burden of high inflation. Lower-income families expend a greater share of their income on necessities; have smaller financial cushions; and may have less ability to switch to lower-priced alternatives" (Brainard 2022). Our research sheds light on these heterogeneous effects and supports the operations management community's ability to mitigate them by providing evidence of the underlying operational mechanisms.

Our findings can inform the responses of operations managers and policymakers to hurricanes. As the federal agency that administers government disaster support services, FEMA is responsible for providing aid to all communities that are declared a federal disaster zone. Its existing processes, however, account for the first-order impacts of a hurricane—supporting the repair and replacement of property, establishing emergency shelter and aid, and replacing damaged food. Those processes do not address the burden of higher prices that we show disproportionately impact low-income communities well beyond a community's immediate recovery phase. Our results can help government agencies account for heterogeneous disaster effects when deploying relief resources. Assistance programs can build off related policy successes. For instance, C&V programs specifically for housing have already been deployed in some settings,[5] and can provide decision autonomy to the victims—the lowest, and arguably most salient, level in an emergency response.

Other potential policy implications of our research are manifold, and require careful consideration and testing. For instance, local authorities can consider prioritizing road clearance and debris removal to facilitate the resupply of grocery stores in all communities. Other critical service infrastructures, such as hospitals, already receive prioritization for road clearance after weather-related events. Constraints on resupplying grocery stores may contribute to the greater prevalence of stockouts in low-income communities. As a retail executive shared with the authors, grocery replenishment after hurricanes may be opportunistic based on which roads are clear, and lower-income communities may not have the resources or the prioritization from shared services to clear roads expediently so truck deliveries to local stores can resume. Resupply may be more responsive in high-income communities simply because of better infrastructure there.

To refine their disaster response efforts, service providers and government agencies can take advantage of their access to spending data similar to IRI, and the increasing quality of analytical methods. This can improve demand forecasting surrounding a hurricane. As our research demonstrates, it is reasonably straightforward to track the prices that consumers are paying in the affected communities, and determine how buying patterns and pricing may be differentially affecting those communities. Data are clearly available in near-real time and in greater detail from the grocery store chains that collect the data and provide them to data aggregators, such as IRI and NielsenIQ. This implies that those chains could provide these data to emergency response agencies, and utilize their own data to ensure that their stores are taking appropriate operational steps to reduce inequities among the communities that they serve.

We believe that our results are generalizable outside of this set of hurricanes. The hurricanes we study (1) impact different geographies, (2) occur in different years, and (3) yield variation in the coefficient estimates that is on par with the variation in the direct damages caused by each hurricane. Despite these differences, there is a remarkable consistency in the sign and statistical significance of our results across hurricanes and postperiods. We note that the largest impact on pricing in low-income communities is associated with Hurricane Katrina, which is distinguished as the only North Atlantic hurricane thus far to exceed $200B in estimated damages. Since 2017, however, three hurricanes passed the $100B threshold—Harvey ($160B), Ian ($119B), and Maria ($115B)—and three others exceeded $50B—Ida ($85B), Helene ($79B), and Irma ($64B) (adjusted to 2024 dollars (NOAA 2024)). Large, damaging hurricanes are unfortunately not as rare as they once were. We hope that our research will motivate future work on operational responses to natural disasters that account for the differential impact on paid prices across disaster-affected communities, and help to alleviate the problems faced by those most in need.

Endnotes

[1] According to the Congressional Budget Office (https://www.cbo.gov/publication/43175), average monthly participation in SNAP has varied between 26 million and 45 million people over our study period. Prior to 2008, and overlapping with our study period, SNAP was known as “food stamps.”

[2] We exclude communities that are also declared disaster zones because of another large hurricane in close temporal proximity to the events we study. Specifically, we exclude communities from our Hurricane Katrina sample that were also declared disaster zones following Hurricane Rita, and we remove communities from our Hurricane Ike sample that were also declared disaster zones following Hurricane Gustav. These exclusions cover both treated and prospective control observations. We run robustness checks without these exclusions, and our results are moderately stronger in magnitude and significance (results available from the authors).

[3] Prior to Hurricane Ike there is an unusual occurrence of products with price points lower than $0.35. These items make up less than 1% of all observations. To identify regularly selling products and train our forecast model, we drop these observations.

[4] There is one affected store in the Katrina analysis that does not report data to IRI in every week before our study period. We remove that store from the analysis because we cannot identify its set of regularly selling products.

[5] For example, the U.S. Department of Housing and Urban Development offers a disaster voucher program to support housing-related costs, https://www.hud.gov/program_offices/public_indian_housing/publications/dvp.

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